

## SAFETY DATA SHEET

Creation Date 01-May-2012

Revision Date 25-Dec-2021

Revision Number 5

### 1. Identification

**Product Name** Aluminium sulfate octadecahydrate

**Cat No. :** AC400590000; AC400590025; AC400590050; AC400595000

**CAS No** 7784-31-8  
**Synonyms** Sulfuric acid, dialuminum salt, octadecahydrate.

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Corrosive to metals  
Serious Eye Damage/Eye Irritation

Category 1  
Category 1

#### Label Elements

**Signal Word**  
Danger

**Hazard Statements**  
May be corrosive to metals  
Causes serious eye damage

**Precautionary Statements****Prevention**

Wear protective gloves/protective clothing/eye protection/face protection

Keep only in original container

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Immediately call a POISON CENTER or doctor/physician

**Spills**

Absorb spillage to prevent material damage

**Storage**

Store in corrosive resistant polypropylene container with a resistant inliner

**Hazards not otherwise classified (HNOC)**

None identified

### 3. Composition/Information on Ingredients

| Component                         | CAS No     | Weight % |
|-----------------------------------|------------|----------|
| Aluminium sulfate octadecahydrate | 7784-31-8  | 100      |
| Aluminum sulfate                  | 10043-01-3 | -        |

### 4. First-aid measures

|                                            |                                                                                                                   |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                            |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.             |
| <b>Most important symptoms and effects</b> | Causes severe eye damage.                                                                                         |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                             |

### 5. Fire-fighting measures

|                                       |                                                                              |
|---------------------------------------|------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>   | Water spray. Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Chemical foam. |
| <b>Unsuitable Extinguishing Media</b> | No information available                                                     |
| <b>Flash Point</b>                    | No information available                                                     |
| <b>Method -</b>                       | No information available                                                     |

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Autoignition Temperature</b>         | No information available |
| <b>Explosion Limits</b>                 |                          |
| <b>Upper</b>                            | No data available        |
| <b>Lower</b>                            | No data available        |
| <b>Sensitivity to Mechanical Impact</b> | No information available |
| <b>Sensitivity to Static Discharge</b>  | No information available |

**Specific Hazards Arising from the Chemical**

Keep product and empty container away from heat and sources of ignition.

**Hazardous Combustion Products**

Sulfur oxides. Burning produces obnoxious and toxic fumes.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
2

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                  |                                                                                                   |
|----------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation. |
| <b>Environmental Precautions</b> | Should not be released into the environment.                                                      |

|                                             |                                                                                                              |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal. |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------|

## 7. Handling and storage

|                 |                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Ensure adequate ventilation. Wear personal protective equipment/face protection. Avoid dust formation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. |
| <b>Storage.</b> | Keep container tightly closed. Keep in a dry, cool and well-ventilated place. Incompatible Materials. Strong oxidizing agents.                                                      |

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component                         | ACGIH TLV | OSHA PEL                           | NIOSH IDLH               | Mexico OEL (TWA) |
|-----------------------------------|-----------|------------------------------------|--------------------------|------------------|
| Aluminium sulfate octadecahydrate |           | (Vacated) TWA: 2 mg/m <sup>3</sup> | TWA: 2 mg/m <sup>3</sup> |                  |
| Aluminum sulfate                  |           | (Vacated) TWA: 2 mg/m <sup>3</sup> | TWA: 2 mg/m <sup>3</sup> |                  |

Legend

OSHA - Occupational Safety and Health Administration

NIOSH IDLH: NIOSH - National Institute for Occupational Safety and Health

|                             |                                                                                                                                                   |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Engineering Measures</b> | Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|

**Personal Protective Equipment**

|                            |                                                                                                                                                                             |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye/face Protection</b> | Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166. |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                 |                                                                                                                                                                                                                                                   |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b> | Wear appropriate protective gloves and clothing to prevent skin exposure.                                                                                                                                                                         |
| <b>Respiratory Protection</b>   | Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. |
| <b>Hygiene Measures</b>         | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                            |

## 9. Physical and chemical properties

|                                               |                                                                      |
|-----------------------------------------------|----------------------------------------------------------------------|
| <b>Physical State</b>                         | Powder Solid                                                         |
| <b>Appearance</b>                             | White                                                                |
| <b>Odor</b>                                   | No information available                                             |
| <b>Odor Threshold</b>                         | No information available                                             |
| <b>pH</b>                                     | 3-4 2% aq.solution                                                   |
| <b>Melting Point/Range</b>                    | 86 °C / 186.8 °F                                                     |
| <b>Boiling Point/Range</b>                    | No information available                                             |
| <b>Flash Point</b>                            | No information available                                             |
| <b>Evaporation Rate</b>                       | Not applicable                                                       |
| <b>Flammability (solid,gas)</b>               | No information available                                             |
| <b>Flammability or explosive limits</b>       |                                                                      |
| Upper                                         | No data available                                                    |
| Lower                                         | No data available                                                    |
| <b>Vapor Pressure</b>                         | No information available                                             |
| <b>Vapor Density</b>                          | Not applicable                                                       |
| <b>Specific Gravity</b>                       | No information available                                             |
| <b>Solubility</b>                             | Soluble in water                                                     |
| <b>Partition coefficient; n-octanol/water</b> | No data available                                                    |
| <b>Autoignition Temperature</b>               | No information available                                             |
| <b>Decomposition Temperature</b>              | No information available                                             |
| <b>Viscosity</b>                              | Not applicable                                                       |
| <b>Molecular Formula</b>                      | Al <sub>2</sub> O <sub>12</sub> S <sub>3</sub> . 18 H <sub>2</sub> O |
| <b>Molecular Weight</b>                       | 666.4                                                                |

## 10. Stability and reactivity

|                                         |                                                           |
|-----------------------------------------|-----------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                |
| <b>Stability</b>                        | Stable under normal conditions.                           |
| <b>Conditions to Avoid</b>              | Heat, flames and sparks. Incompatible products.           |
| <b>Incompatible Materials</b>           | Strong oxidizing agents                                   |
| <b>Hazardous Decomposition Products</b> | Sulfur oxides, Burning produces obnoxious and toxic fumes |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                  |
| <b>Hazardous Reactions</b>              | None under normal processing.                             |

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component        | LD50 Oral           | LD50 Dermal       | LC50 Inhalation |
|------------------|---------------------|-------------------|-----------------|
| Aluminum sulfate | 6207 mg/kg ( Mouse) | >5 g/kg ( Rabbit) | Not listed      |

**Toxicologically Synergistic Products** No information available

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Irritation** Risk of serious damage to eyes

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                         | CAS No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------------------------------|------------|------------|------------|------------|------------|------------|
| Aluminium sulfate octadecahydrate | 7784-31-8  | Not listed | Not listed | Not listed | Not listed | Not listed |
| Aluminum sulfate                  | 10043-01-3 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

| Component                         | Freshwater Algae | Freshwater Fish                                     | Microtox                                                                                                                           | Water Flea                                 |
|-----------------------------------|------------------|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| Aluminium sulfate octadecahydrate | -                | -                                                   | EC50 = 1.04 mg/L 30 min<br>EC50 = 1.08 mg/L 20 min<br>EC50 = 1.10 mg/L 15 min<br>EC50 = 1.28 mg/L 10 min<br>EC50 = 1.62 mg/L 5 min | -                                          |
| Aluminum sulfate                  | Not listed       | LC50: = 27.9 mg/L, 96h static (Pimephales promelas) | Not listed                                                                                                                         | 136 mg/L EC50 15 min<br>38.2 mg/L EC50 48h |

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No

UN3260

Proper Shipping Name

CORROSIVE SOLID, ACIDIC, INORGANIC, N.O.S.

|                             |                                            |
|-----------------------------|--------------------------------------------|
| <b>Technical Name</b>       | Aluminium sulfate octadecahydrate          |
| <b>Hazard Class</b>         | 8                                          |
| <b>Packing Group</b>        | III                                        |
| <b>TDG</b>                  |                                            |
| <b>UN-No</b>                | UN3260                                     |
| <b>Proper Shipping Name</b> | CORROSIVE SOLID, ACIDIC, INORGANIC, N.O.S. |
| <b>Hazard Class</b>         | 8                                          |
| <b>Packing Group</b>        | III                                        |
| <b>IATA</b>                 |                                            |
| <b>UN-No</b>                | UN3260                                     |
| <b>Proper Shipping Name</b> | CORROSIVE SOLID, ACIDIC, INORGANIC, N.O.S. |
| <b>Hazard Class</b>         | 8                                          |
| <b>Packing Group</b>        | III                                        |
| <b>IMDG/IMO</b>             |                                            |
| <b>UN-No</b>                | UN3260                                     |
| <b>Proper Shipping Name</b> | CORROSIVE SOLID, ACIDIC, INORGANIC, N.O.S. |
| <b>Hazard Class</b>         | 8                                          |
| <b>Packing Group</b>        | III                                        |

## 15. Regulatory information

### United States of America Inventory

| Component                         | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|-----------------------------------|------------|------|-----------------------------------------------|-----------------------------|
| Aluminium sulfate octadecahydrate | 7784-31-8  | -    | -                                             | -                           |
| Aluminum sulfate                  | 10043-01-3 | X    | ACTIVE                                        | -                           |

#### Legend:

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA 12(b)** - Notices of Export Not applicable

### International Inventories

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                         | CAS No     | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|-----------------------------------|------------|-----|------|-----------|-------|------|------|------|-------|----------|
| Aluminium sulfate octadecahydrate | 7784-31-8  | -   | -    | -         | -     | -    |      | X    | X     | -        |
| Aluminum sulfate                  | 10043-01-3 | X   | -    | 233-135-0 | X     | X    | X    | X    | X     | KE-01042 |

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### U.S. Federal Regulations

**SARA 313** Not applicable

**SARA 311/312 Hazard Categories** See section 2 for more information

### CWA (Clean Water Act)

| Component        | CWA - Hazardous Substances | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants |
|------------------|----------------------------|-----------------------------|------------------------|---------------------------|
| Aluminum sulfate | X                          | 5000 lb                     | -                      | -                         |

**Clean Air Act** Not applicable

**OSHA** - Occupational Safety and Not applicable

Health Administration

**CERCLA**

| Component        | Hazardous Substances RQs | CERCLA EHS RQs |
|------------------|--------------------------|----------------|
| Aluminum sulfate | 5000 lb                  | -              |

**California Proposition 65**

This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component                         | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|-----------------------------------|---------------|------------|--------------|----------|--------------|
| Aluminium sulfate octadecahydrate | -             | -          | X            | -        | -            |
| Aluminum sulfate                  | X             | X          | X            | -        | -            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): Y  
 DOT Marine Pollutant N  
 DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security**

This product does not contain any DHS chemicals.

**Other International Regulations**

**Mexico - Grade** No information available

**Authorisation/Restrictions according to EU REACH****Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                         | CAS No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Aluminium sulfate octadecahydrate | 7784-31-8  | Not applicable | Not applicable               | Not applicable            | Not applicable                             |
| Aluminum sulfate                  | 10043-01-3 | Listed         | Not applicable               | Not applicable            | Not applicable                             |

| Component                         | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Aluminium sulfate octadecahydrate | 7784-31-8  | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Aluminum sulfate                  | 10043-01-3 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information****Prepared By**

Regulatory Affairs  
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**Revision Summary**

This document has been updated to comply with the US OSHA HazCom 2012 Standard

replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS).

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**